

Администрирование сетевых подсистем

Настройка HTTP-сервера Apache (Лабораторная работа №4)

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Цели и задачи работы

Приобретение практических навыков по установке и базовому конфигурированию HTTP-сервера Apache, а также настройке виртуального хостинга.

Выполнение лабораторной работы

Установка HTTP-сервера

```
root@server:~# sudo -i

(14/14): httpd-manual-2.4.63-1.el10_0.2.noarch.rpm                41 MB/s | 2.3 MB   00:00
-----
Total                                                            13 MB/s | 4.6 MB   00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing :                                                    1/1
  Running scriptlet: httpd-filesystem-2.4.63-1.el10_0.2.noarch 1/14
  Installing : httpd-filesystem-2.4.63-1.el10_0.2.noarch      1/14
  Installing : apr-1.7.5-2.el10.x86_64                        2/14
  Installing : apr-util-ldap-1.6.3-21.el10.x86_64             3/14
  Installing : apr-util-openssl-1.6.3-21.el10.x86_64          4/14
  Installing : apr-util-1.6.3-21.el10.x86_64                  5/14
  Installing : httpd-tools-2.4.63-1.el10_0.2.x86_64           6/14
  Installing : httpd-core-2.4.63-1.el10_0.2.x86_64            7/14
  Installing : mod_http2-2.0.29-2.el10_0.1.x86_64             8/14
  Installing : mod_lua-2.4.63-1.el10_0.2.x86_64               9/14
  Installing : rocky-logos-httpd-100.4-7.el10.noarch          10/14
  Installing : httpd-2.4.63-1.el10_0.2.x86_64                 11/14
  Running scriptlet: httpd-2.4.63-1.el10_0.2.x86_64           11/14
  Installing : httpd-manual-2.4.63-1.el10_0.2.noarch          12/14
  Installing : mod_fcgid-2.3.9-35.el10.x86_64                 13/14
  Installing : mod_ssl-1.2.4.63-1.el10_0.2.x86_64             14/14
  Running scriptlet: httpd-2.4.63-1.el10_0.2.x86_64          14/14
  Running scriptlet: mod_ssl-1.2.4.63-1.el10_0.2.x86_64      14/14

Installed:
  apr-1.7.5-2.el10.x86_64          apr-util-1.6.3-21.el10.x86_64          apr-util-ldap-1.6.3-21.el10.x86_64
  apr-util-openssl-1.6.3-21.el10.x86_64  httpd-2.4.63-1.el10_0.2.x86_64  httpd-core-2.4.63-1.el10_0.2.x86_64
  httpd-filesystem-2.4.63-1.el10_0.2.noarch  httpd-manual-2.4.63-1.el10_0.2.noarch  httpd-tools-2.4.63-1.el10_0.2.x86_64
  mod_fcgid-2.3.9-35.el10.x86_64          mod_http2-2.0.29-2.el10_0.1.x86_64  mod_lua-2.4.63-1.el10_0.2.x86_64
  mod_ssl-1.2.4.63-1.el10_0.2.x86_64      rocky-logos-httpd-100.4-7.el10.noarch

Complete!
[root@server.manovichkov.net ~]#
```

Рис. 1:

Установка пакетов httpd

Настройка firewall и запуск сервиса

```
ient ganglia-master git gpssd grafana gre high-availability http http3 https ident imap imaps ipert2 ipert3 ipfs ipp ipp-cl
ient ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-
apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodepor
t-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt lib
virt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp
mongodb mosh mountd mpd mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios-ns netdata-
dashboard nfs nfs3 nmap-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pm
cd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3nets
rv ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-
master samba samba-client samba-dc sane settlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp s
nmp-tls snmp-tls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh statsrv steam-lan-transfer steam-streaming stel
laris stronghold-crusader stun stuns submission supertuxkart svdrp svn syncthing syncthing-gui syncthing-relay synergy sys
comlan syslog syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks transmission-client turn turns upnp-client vd
sm vnc-server vrrp warpinator wben-http wben-https wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discove
ry-tcp ws-discovery-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix
-java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier
[root@server.manovichkov.net ~]# firewall-cmd --add-service=http
success
[root@server.manovichkov.net ~]# firewall-cmd --add-service=http --permanent
success
[root@server.manovichkov.net ~]# systemctl enable httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.
[root@server.manovichkov.net ~]# systemctl start httpd
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]#
```

Рис. 2: Настройка firewall для http

Проверка работы сервера

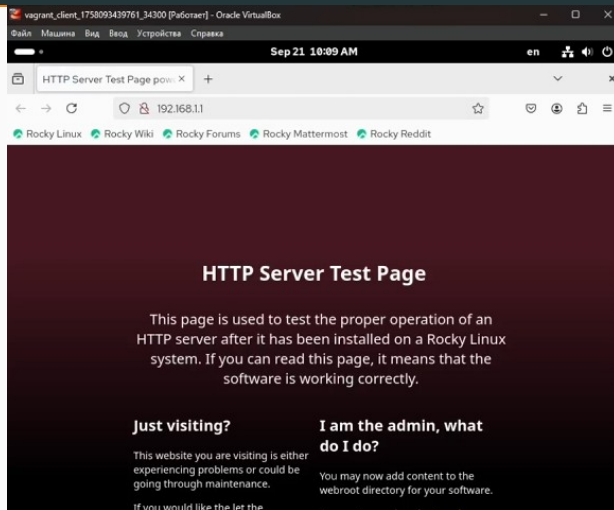


Рис. 3:

Анализ логов работы сервера

```
[root@server.manovichkov.net ~]# tail -f /var/log/httpd/error_log
[Sun Sep 21 10:05:08.130316 2025] [suexec:notice] [pid 16550:tid 16550] AH01232: suEXEC mechanism enabled (wrapper: /usr/s
bin/suexec)
[Sun Sep 21 10:05:08.139142 2025] [lbmethod_heartbeat:notice] [pid 16550:tid 16550] AH02282: No slotmem from mod_heartmoni
tor
[Sun Sep 21 10:05:08.140319 2025] [systemd:notice] [pid 16550:tid 16550] SELinux policy enabled; httpd running as context
system_u:system_r:httpd_t:s0
[Sun Sep 21 10:05:08.142119 2025] [mpm_event:notice] [pid 16550:tid 16550] AH00489: Apache/2.4.63 (Rocky Linux) OpenSSL/3.
2.2 mod_fcgid/2.3.9 configured -- resuming normal operations
[Sun Sep 21 10:05:08.142129 2025] [core:notice] [pid 16550:tid 16550] AH00094: Command line: '/usr/sbin/httpd -D FOREGROUN
D'
[Sun Sep 21 10:09:32.145373 2025] [autoindex:error] [pid 16554:tid 16705] [client 192.168.1.30:55606] AH01276: Cannot serv
e directory /var/www/html/: No matching DirectoryIndex (index.html) found, and server-generated directory index forbidden
by Options directive
^C
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# tail -f /var/log/httpd/access_log
192.168.1.30 - - [21/Sep/2025:10:09:32 +0000] "GET / HTTP/1.1" 403 7620 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gec
ko/20100101 Firefox/128.0"
192.168.1.30 - - [21/Sep/2025:10:09:32 +0000] "GET /icons/poweredby.png HTTP/1.1" 200 15443 "http://192.168.1.1/" "Mozilla
/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0"
192.168.1.30 - - [21/Sep/2025:10:09:32 +0000] "GET /poweredby.png HTTP/1.1" 200 5714 "http://192.168.1.1/" "Mozilla/5.0 (X
11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0"
192.168.1.30 - - [21/Sep/2025:10:09:32 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http://192.168.1.1/" "Mozilla/5.0 (X11;
Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0"
```

Рис. 4:

Анализ логов веб-сервера


```

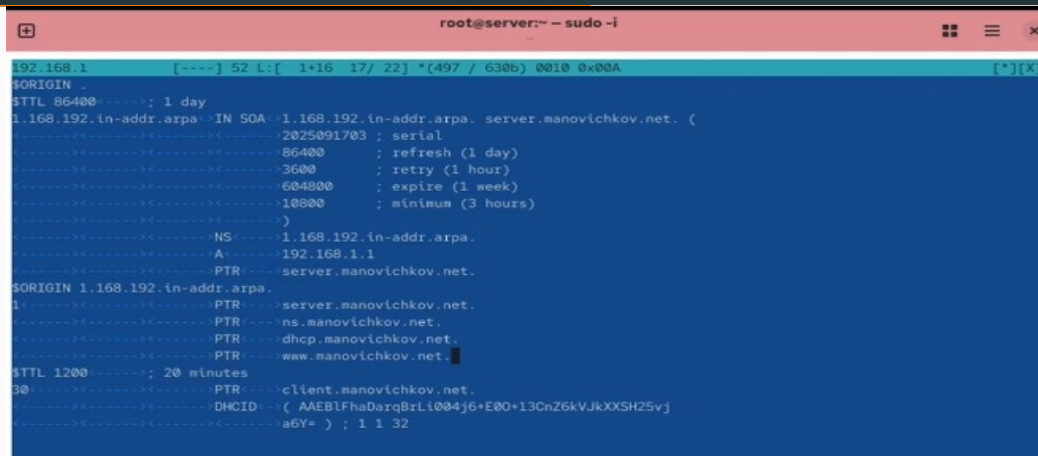
root@server:~ - sudo -i

manovichkov.net  [-M--] 43 L:[ 1+20 21/ 21] *(548 / 548b) <EOF>
$ORIGIN .
$TTL 86400<----->; 1 day
manovichkov.net<----->IN SOA<----->manovichkov.net. (
<-----><-----><-----><----->2025091704 ; serial
<-----><-----><-----><----->86400      ; refresh (1 day)
<-----><-----><-----><----->3600       ; retry (1 hour)
<-----><-----><-----><----->604800     ; expire (1 week)
<-----><-----><-----><----->10800      ; minimum (3 hours)
<-----><-----><-----><----->)
<-----><-----><-----><----->NS<----->manovichkov.net.
<-----><-----><-----><----->A<----->192.168.1.1
$ORIGIN manovichkov.net.
$TTL 1200<----->; 20 minutes
client<-----><-----><-----><----->A<----->192.168.1.30
<-----><-----><-----><----->DHCHID<-->( AAEB1FhaDarqBrLi004j6+E00+13CnZ6kVJkXXSH25vj
<-----><-----><-----><----->a6Y= ) ; 1 1 32
$TTL 86400<----->; 1 day
dhcp<-----><-----><-----><----->A<----->192.168.1.1
ns<-----><-----><-----><----->A<----->192.168.1.1
server<-----><-----><-----><----->A<----->192.168.1.1
www<-----><-----><-----><----->A<----->192.168.1.1

```

Рис. 5:

Изменения в DNS-зоне

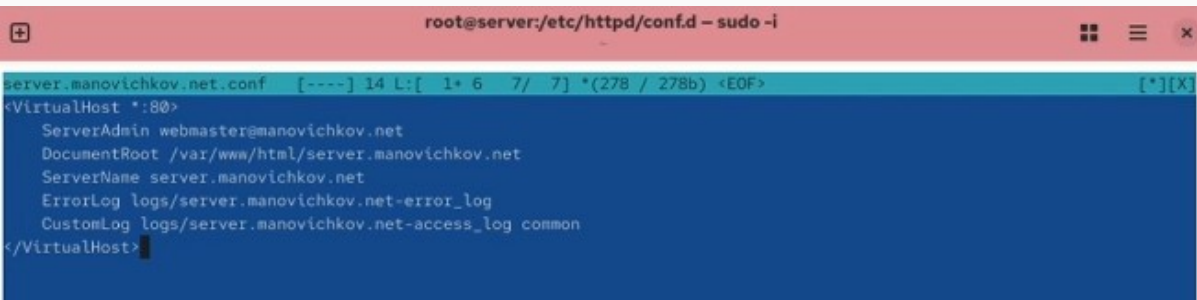


```
root@server:~ - sudo -i
192.168.1.1 [----] 52 L: [ 1+16 17/ 22] *(497 / 630b) 0010 0x00A [*][X]
$ORIGIN .
$TTL 86400 : 1 day
1.168.192.in-addr.arpa IN SOA 1.168.192.in-addr.arpa. server.manovichkov.net. (
    2025091703 ; serial
    86400      ; refresh (1 day)
    3600      ; retry (1 hour)
    604800    ; expire (1 week)
    10800     ; minimum (3 hours)
)
; NS
; A
; PTR
$ORIGIN 1.168.192.in-addr.arpa.
1 PTR server.manovichkov.net.
; PTR
; PTR
; PTR
$TTL 1200 : 20 minutes
30 PTR client.manovichkov.net.
; DHCID
; a6Y = ) ; 1 1 32
```

Рис. 6:

Настройка записей в обратной зоне

Конфигурация виртуальных хостов



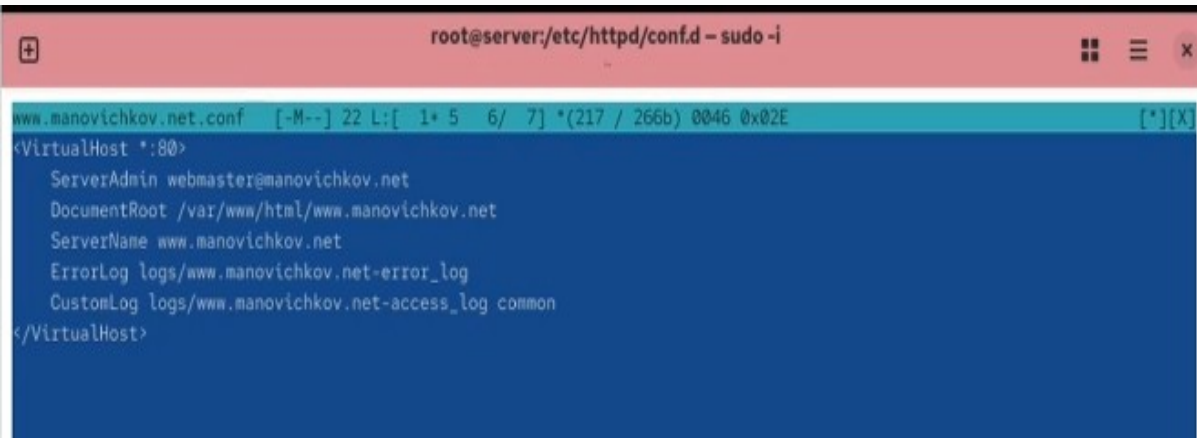
The image shows a terminal window with a red title bar. The title bar text is "root@server:/etc/httpd/conf.d - sudo -i". On the right side of the title bar are three icons: a window icon, a hamburger menu icon, and a close icon. The terminal content shows a file editor interface for "server.manovichkov.net.conf". The top status bar of the editor displays "[----] 14 L:[1+ 6 7/ 7] *(278 / 278b) <EOF> [*][X]". The main editing area has a dark blue background and contains the following XML configuration for a VirtualHost:

```
<VirtualHost *:80>
    ServerAdmin webmaster@manovichkov.net
    DocumentRoot /var/www/html/server.manovichkov.net
    ServerName server.manovichkov.net
    ErrorLog logs/server.manovichkov.net-error_log
    CustomLog logs/server.manovichkov.net-access_log common
</VirtualHost>
```

Рис. 7: Конфигурация server.manovichkov.net

Автоматизация настройки

Конфигурация виртуальных хостов



The image shows a terminal window with a pink title bar. The title bar text is "root@server:/etc/httpd/conf.d - sudo -i". On the right side of the title bar are three icons: a window icon, a hamburger menu icon, and a close button icon. The terminal content shows a file named "www.manovichkov.net.conf" with a status bar at the top indicating it is 22 lines long, contains 5 tabs, and is 266 bytes in size. The main content of the file is an XML configuration for a VirtualHost on port 80. The configuration includes the following settings:

```
<VirtualHost *:80>  
    ServerAdmin webmaster@manovichkov.net  
    DocumentRoot /var/www/html/www.manovichkov.net  
    ServerName www.manovichkov.net  
    ErrorLog logs/www.manovichkov.net-error_log  
    CustomLog logs/www.manovichkov.net-access_log common  
</VirtualHost>
```

Рис. 8: Конфигурация www.manovichkov.net

Создание веб-контента

```
[root@server.manovichkov.net conf.d]# cd /var/www/html
[root@server.manovichkov.net html]# mkdir server.manovichkov.net
[root@server.manovichkov.net html]# cd server.manovichkov.net/
[root@server.manovichkov.net server.manovichkov.net]# touch index.html
[root@server.manovichkov.net server.manovichkov.net]# mcedit index.html

[root@server.manovichkov.net server.manovichkov.net]# cd ..
[root@server.manovichkov.net html]# cd www.manovichkov.net/
-bash: cd: www.manovichkov.net/: No such file or directory
[root@server.manovichkov.net html]# mkdir www.manovichkov.net/
[root@server.manovichkov.net html]# cd www.manovichkov.net/
[root@server.manovichkov.net www.manovichkov.net]# touch index.html
[root@server.manovichkov.net www.manovichkov.net]# mcedit index.html

[root@server.manovichkov.net www.manovichkov.net]#
[root@server.manovichkov.net www.manovichkov.net]# chown -R apache:apache /var/www
[root@server.manovichkov.net www.manovichkov.net]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.manovichkov.net www.manovichkov.net]# restorecon -vR /var/named/
[root@server.manovichkov.net www.manovichkov.net]# restorecon -vR /var/www
[root@server.manovichkov.net www.manovichkov.net]# systemctl restart httpd
```

Рис. 9: Создание тестовых страниц

Проверка работы виртуальных хостов

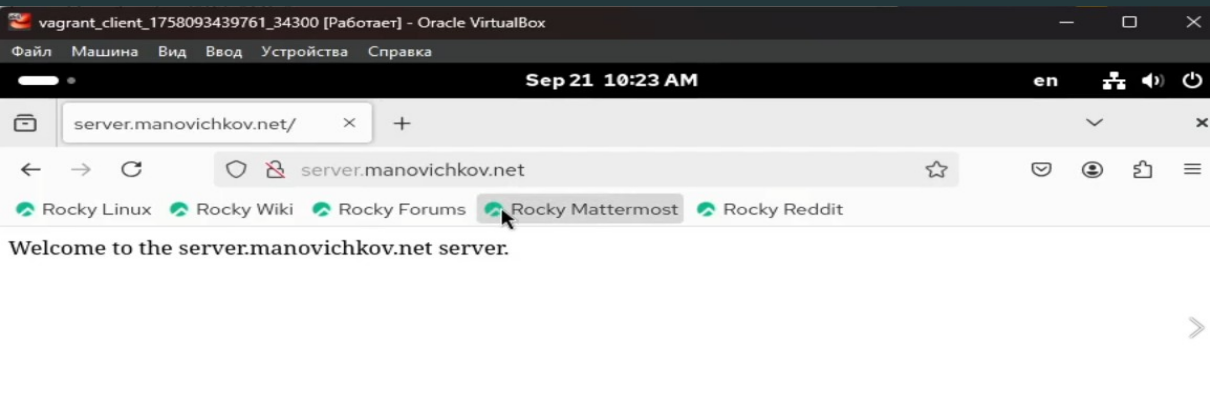


Рис. 10: Проверка сайта server.manovichkov.net

Проверка работы виртуальных хостов

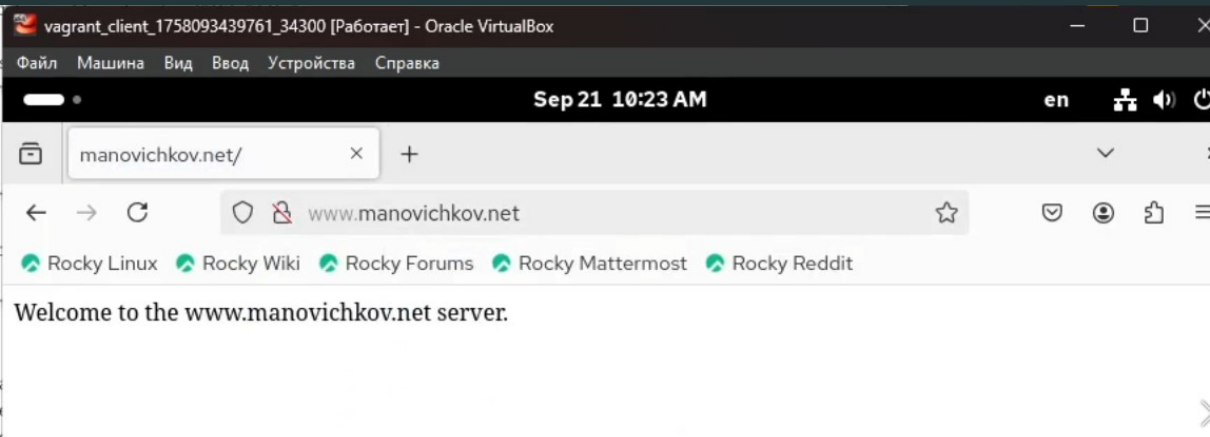


Рис. 11: Проверка сайта www.manovichkov.net

Выводы по проделанной работе

В ходе лабораторной работы был установлен и настроен веб-сервер Apache.

Созданы виртуальные хосты с отдельными доменными именами, произведена интеграция с DNS и добавлены тестовые страницы.

Сконфигурирована автоматизация с помощью скриптов Vagrant, что упрощает повторное развёртывание и настройку системы.